

Irreducibility in $\mathbb{K}[x]$

- **Definition 1**

A polynomial $p(x) \in \mathbb{K}[x]$ with $\deg(p(x)) > 0$ is **irreducible** over $\mathbb{K}$ if whenever $p(x) = q(x)h(x)$ for some $q(x), h(x) \in \mathbb{K}[x]$ then either $q(x)$ or $h(x)$ is a constant polynomial. Otherwise, $p(x)$ is **reducible**.

- **Example 1.1**

The polynomial $p(x) = x^2 - 3$ is irreducible over $\mathbb{Q}$ but is reducible over $\mathbb{R}$ since

$$p(x) = (x + \sqrt{3})(x - \sqrt{3}).$$

- **Definition 2**

Let $p(x) \in \mathbb{K}[x]$. An element $\alpha \in \mathbb{K}$ is a root of $p(x)$ if $p(\alpha) = 0$.

- **Example 2.1**

The polynomial $p(x) = (x - 1)(x - 2)(x + 3)$ has roots 1, 2 and -3 .

The polynomial $p(x) = x^2 + 1$ has no roots in $\mathbb{R}$ but has two roots i and $-i$ in $\mathbb{C}$.

- **Definition 3**

Let $p(x), q(x) \in \mathbb{K}[x]$. We say that $q(x)$ divides $p(x)$, denoted $q(x) \mid p(x)$, if there exists $h(x) \in \mathbb{K}[x]$ such that

$$p(x) = q(x)h(x).$$

- **Theorem 3.1 (Factor Theorem)**

Let $p(x) \in \mathbb{K}[x]$ and $\alpha \in \mathbb{K}$, then α is a root of $p(x)$ if and only if $(x - \alpha) \mid p(x)$.

- **Proof 3.1.1**

Suppose $p(x) \in \mathbb{K}[x]$ and $\alpha \in \mathbb{K}$.

($\implies$) Suppose that α is a root of $p(x)$ then $p(\alpha) = 0$ and by the remainder theorem there exists $q(x) \in \mathbb{K}[x]$ such that

$$p(x) = (x - \alpha)q(x) + p(\alpha) = (x - \alpha)q(x).$$

That is, $(x - \alpha) \mid p(x)$.

($\impliedby$) Suppose that $(x - \alpha) \mid p(x)$. By definition, there exists $q(x) \in \mathbb{K}[x]$ such that $p(x) = (x - \alpha)q(x)$. Hence $p(\alpha) = 0$ and thus α is a root of $p(x)$.

- **Example 3.2**

Consider $p(x) = x^3 - 2x^2 - 5x + 6$. Since $p(1) = 1 - 2 - 5 + 6 = 0$ then by the factor theorem $(x - 1) \mid p(x)$. Furthermore

$$p(x) = (x - 1)(x^2 - x - 6).$$

- **Corollary 3.3**

Let $p(x) \in \mathbb{K}[x]$ with $\deg(p(x)) > 1$. If $p(x)$ has a root in $\mathbb{K}$ then $p(x)$ is reducible over $\mathbb{K}$.

- **Proof 3.3.1**

Let α be a root of $p(x)$, then by the factor theorem $(x - \alpha) \mid p(x)$, that is, there exists a polynomial $q(x) \in \mathbb{K}[x]$ such that $p(x) = (x - \alpha)q(x)$. Since

$$\deg(p(x)) = \deg((x - \alpha)q(x)) = 1 + \deg(q(x)) > 1.$$

Then $q(x)$ is not a constant polynomial and thus $p(x)$ is reducible over $\mathbb{K}$.

- **Remark 3.4**

The converse is false. Consider $p(x) = x^4 + 2x^2 + 1 \in \mathbb{R}[x]$, then $p(x)$ is reducible over $\mathbb{R}$ since

$$p(x) = (x^2 + 1)(x^2 + 1).$$

But $p(x)$ has no roots in $\mathbb{R}$.

◦ **Proposition 3.5**

Let $p(x) \in \mathbb{K}[x]$.

1. If $\deg(p(x)) = 1$, then $p(x)$ is irreducible over $\mathbb{K}$.
2. If $\deg(p(x))$ is 2 or 3, then $p(x)$ is irreducible over $\mathbb{K}$ if and only if $p(x)$ has no roots in $\mathbb{K}$.

■ **Proof 3.5.1**

1. If $p(x) = q(x)h(x)$ then

$$1 = \deg(p(x)) = \deg(q(x)h(x)) = \deg(q(x)) + \deg(h(x)).$$

Hence $\deg(q(x)) = 0 \vee \deg(h(x)) = 0$, that is, $q(x)$ is a constant polynomial or $h(x)$ is a constant polynomial. Therefore $p(x)$ is irreducible over $\mathbb{K}$.

2. ($\implies$) Suppose that $p(x)$ is irreducible over $\mathbb{K}$ then by the corollary $p(x)$ has no roots in $\mathbb{K}$.

($\impliedby$) Suppose that $p(x)$ has no roots in $\mathbb{K}$. If $p(x)$ is reducible over $\mathbb{K}$ then $p(x) = q(x)h(x)$ for some nonconstant polynomials $q(x), h(x) \in \mathbb{K}[x]$.

Hence

$$2 = \deg(p(x)) = \deg(q(x)h(x)) = \deg(q(x)) + \deg(h(x)).$$

Then $\deg(q(x)) = \deg(h(x)) = 1$, that is, $q(x) = ax + b$ with $a, b \in \mathbb{K}$ and $a \neq 0$ with $q(-a^{-1}b) = 0$ and $-a^{-1}b \in \mathbb{K}$. Therefore $-a^{-1}b$ is a root of $p(x)$ which is a contradiction.

Similarly if $\deg(p(x)) = 3$. Thus $p(x)$ is irreducible over $\mathbb{K}$.

◦ **Theorem 3.6**

Let $p(x) \in \mathbb{K}[x]$ with $p(x) \neq 0$. If $\deg(p(x)) = n$ then $p(x)$ has at most n roots in $\mathbb{K}$.

■ **Proof 3.6.1**

By induction on n .

1. **Base step** If $n = 1$ then $p(x) = ax + b$ with $a, b \in \mathbb{K}$ and $a \neq 0$ which has exactly one root.
2. **Inductive step** Suppose that the result is true for n and consider $\deg(p(x)) = n + 1$. If $p(x)$ has no roots in $\mathbb{K}$ there is nothing to prove. If $p(x)$ has roots in $\mathbb{K}$, let $\alpha \in \mathbb{K}$ be a root of $p(x)$. Then by the factor theorem $(x - \alpha) \mid p(x)$, that is,

$$p(x) = (x - \alpha)q(x) \quad \text{for some } q(x) \in \mathbb{K}[x].$$

Furthermore

$$n + 1 = \deg(p(x)) = \deg((x - \alpha)q(x)) = 1 + \deg(q(x)).$$

Then $\deg(q(x)) = n$ and by the inductive hypothesis $q(x)$ has at most n roots in $\mathbb{K}$. If $\beta \in \mathbb{K}$ is another root of $p(x)$ then $p(\beta) = (\beta - \alpha)q(\beta) = 0$. So we obtain that $\beta = \alpha$ or $q(\beta) = 0$. Therefore β is among the at most n roots, that is $p(x)$ has at most $n + 1$ roots.

- **Definition 4**

Let $p(x) \in \mathbb{K}[x]$ with $p(x) \neq 0$ and let $\alpha \in \mathbb{K}$ be a root of $p(x)$. The **multiplicity** of α is the largest integer m such that

$$(x - \alpha)^m \mid p(x).$$

- **Remark 4.1**

Equivalently, α has multiplicity m when

$$p(x) = (x - \alpha)^m q(x)$$

with $q(\alpha) \neq 0$.

- **Example 4.2**

If $p(x) = (x - 2)^3(x + 1)$ then 2 is a root of $p(x)$ with multiplicity 3 and -1 is a root of $p(x)$ with multiplicity 1.

- **Theorem 4.3 (Fundamental theorem of algebra)**

If $p(x) \in \mathbb{C}[x]$ with $\deg(p(x)) \geq 1$ then $p(x)$ has a root in $\mathbb{C}$. Equivalently, every polynomial $p(x) \in \mathbb{C}[x]$ has exactly n roots in $\mathbb{C}$ where roots are counted with multiplicities.

- **Corollary 4.4**

A nonconstant polynomial in $\mathbb{C}[x]$ is irreducible if and only if it has degree 1.

- **Corollary 4.5**

Let $p(x) \in \mathbb{C}[x]$ with $\deg(p(x)) = n \geq 1$. Then

$$p(x) = \alpha_0(x - \alpha_1)^{m_1}(x - \alpha_2)^{m_2} \cdots (x - \alpha_s)^{m_s}$$

where $\alpha_1, \alpha_2, \dots, \alpha_s$ are the distinct roots of $p(x)$, $m_1, m_2, \dots, m_s$ are the multiplicities with $m_1 + m_2 + \cdots + m_s = n$.

- **Theorem 4.6**

Let $p(x) \in \mathbb{R}[x]$. If $z = a + bi$ is a root of $p(x)$ then $\bar{z} = a - bi$ is also a root of $p(x)$.

- **Proof 4.6.1**

Let $p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n \in \mathbb{R}[x]$ and let $z = a + bi$ be a root of $p(x)$, that is, $p(z) = 0$. Hence

$$\begin{aligned} p(\bar{z}) &= a_0 + a_1\bar{z} + a_2(\bar{z})^2 + \cdots + a_n(\bar{z})^n \\ &= a_0 + a_1\bar{z} + a_2\bar{z}^2 + \cdots + a_n\bar{z}^n \\ &= \overline{a_0 + a_1z + a_2z^2 + \cdots + a_nz^n} \\ &= \overline{a_0 + a_1z + a_2z^2 + \cdots + a_nz^n} \\ &= \overline{p(z)} = \bar{0} = 0. \end{aligned}$$

- **Corollary 4.7**

Every polynomial $p(x) \in \mathbb{R}[x]$ with odd degree has at least one real root.

- **Proof 4.7.1**

Suppose that $\deg(p(x)) = 2k + 1$ for some $k \in \mathbb{N}_0$ then $p(x)$ has roots $\alpha_1, \alpha_2, \dots, \alpha_{2k+1} \in \mathbb{C}$ and by Theorem 4.6, $\bar{\alpha}_1, \bar{\alpha}_2, \dots, \bar{\alpha}_{2k+1}$ are also roots of $p(x)$. There exist some roots such that $\alpha_i = \bar{\alpha}_i$ and thus $\alpha_i \in \mathbb{R}$.

- **Theorem 4.8 (Rational roots theorem)**

Let $p(x) = a_0 + a_1x + \cdots + a_nx^n \in \mathbb{Z}[x]$ with $a_n \neq 0$ and let $\alpha = \frac{r}{s} \in \mathbb{Q}$ with $r \in \mathbb{Z}$, $s \in \mathbb{Z}$ and r, s are coprime. If α is a root of $p(x)$ then we have that $r \mid a_0$ and $s \mid a_n$. For example, let

$$p(x) = 3x^3 + 2x^2 - 2x - 8.$$

The possible rational roots are

$$\pm 1, \pm 2, \pm 4, \pm 8, \pm \frac{1}{3}, \pm \frac{2}{3}, \pm \frac{4}{3}, \pm \frac{8}{3}.$$